

PROGRAM:

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#include <stdio.h>
#include <stdlib.h>
#define MAX 100 // Maximum size of the queue
// Queue structure definition
struct Queue {
    int arr[MAX];
    int front;
    int rear;
};
// Function prototypes
void initializeQueue(struct Queue *queue);
int isFull(struct Queue *queue);
int isEmpty(struct Queue *queue);
void enqueue(struct Queue *queue, int element);
int dequeue(struct Queue *queue);
void displayQueue(struct Queue *queue);
int main() {
    struct Queue queue;
    int choice, element;
    initializeQueue(&queue);
    while (1) {
        printf("\nQueue Operations Menu:\n");
        printf("1. Enqueue\n");
        printf("2. Dequeue\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter element to enqueue: ");
                scanf("%d", &element);
                enqueue(&queue, element);
                break;
            case 2:
                element = dequeue(&queue);
                if (element != -1)
                    printf("Dequeued element: %d\n", element);
                break;
            case 3:
                displayQueue(&queue);
                break;
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case 4:
exit(0);
default:
printf("Invalid choice! Please enter a valid option.\n");
}
}
return 0;
}
// Function to initialize the queue
void initializeQueue(struct Queue *queue) {
queue->front = -1;
queue->rear = -1;
}
// Function to check if the queue is full
int isFull(struct Queue *queue) {
return queue->rear == MAX - 1;
}
// Function to check if the queue is empty
int isEmpty(struct Queue *queue) {
return queue->front == -1 || queue->front > queue->rear;
}
// Function to enqueue an element to the queue
void enqueue(struct Queue *queue, int element) {
if (isFull(queue)) {
printf("Error: Queue overflow. Cannot enqueue element.\n");
return;
}
if (isEmpty(queue)) {
queue->front = 0;
}
queue->arr[++queue->rear] = element;
printf("Element enqueued: %d\n", element);
}
// Function to dequeue an element from the queue
int dequeue(struct Queue *queue) {
if (isEmpty(queue)) {
printf("Error: Queue underflow. Cannot dequeue element.\n");
return -1;
}
return queue->arr[queue->front++];
}
// Function to display the queue elements
void displayQueue(struct Queue *queue) {
if (isEmpty(queue)) {

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printf("Queue is empty.\n");  
return;  
}  
printf("Queue elements: ");  
for (int i = queue->front; i <= queue->rear; i++) {  
    printf("%d ", queue->arr[i]);  
}  
printf("\n");  
}
```